

----- Active space -----

CONGENITAL ANOMALIES OF KIDNEY AND URINARY TRACT

Renal Agenesis

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UNILATERAL RENAL AGENESIS :

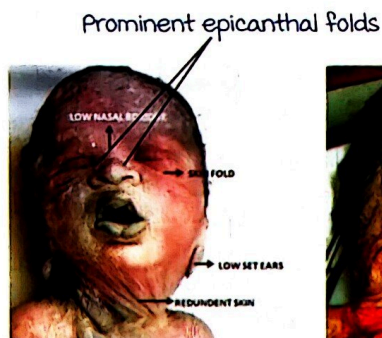
- Normal contralateral kidney → **Hypertrophy** → Normal overall renal function → Patient is asymptomatic.
- **Incidental finding** on USG.
- Treatment : Reassurance, avoid contact sports (Trauma to unaffected kidney).
- Female : Associated **MRKH** (Mayer-Rokitansky-Kuster-Hauser) syndrome.



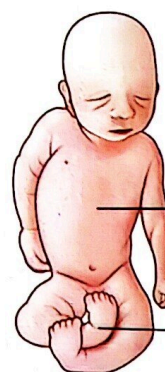
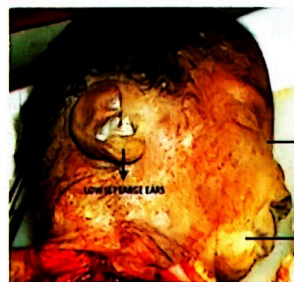
Pelvis USG : Rule out uterine maldevelopment, vaginal aplasia, normal ovaries.

BILATERAL RENAL AGENESIS :

- Both kidneys not developed : Severe presentation.
 - ↓ urine output → ↓ amniotic fluid volume → **Severe oligohydramnios** (Antenatal period)
 - ↓
 - Fetal compression
 - ↓
 - Potter's syndrome**



Facial features



Features in the body

Pulmonary hypoplasia
(d/t chest compression)
If severe → Death

Limb deformities (Eg :
CTEV : Congenital talipes
equino varus)

Renal Dysplasia

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- Dysplasia: Abnormal differentiation
- **m/c : Cystic dysplasia** → m/c : multicystic dysplastic kidney (MCDK) d/t abnormal metanephric differentiation

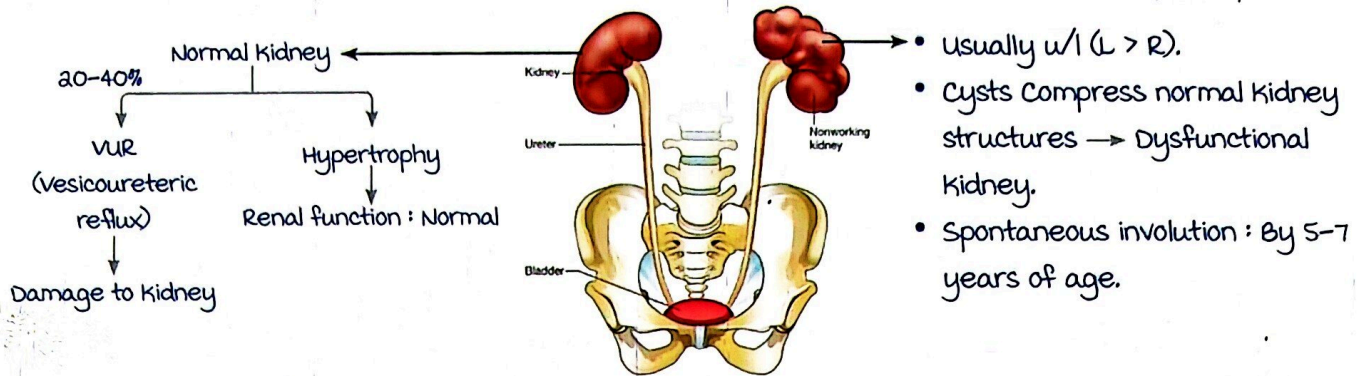


m/c : Cystic renal malformation

Congenital Anomalies of Kidney & Urinary Tract

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PATHOGENESIS :



Autosomal Recessive Polycystic Kidney Disease (ARPKD)

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- Bilateral condition → ↓ Kidney function → Progression to CKD, uncontrolled HTN.
 - Characterized by multiple cysts.
- Gene defect : PKHD1 gene (Chromosome 6p).

Hepatic manifestations : Liver cirrhosis (Periportal fibrosis)
O/E : Bilateral abdominal mass : Loin area.

Diagnosis :

Gene testing : For PKHD1 gene.

USG → Hyperechoic bright kidneys
→ Loss of cortico medullary differentiation



Polycystic Kidney

Treatment :

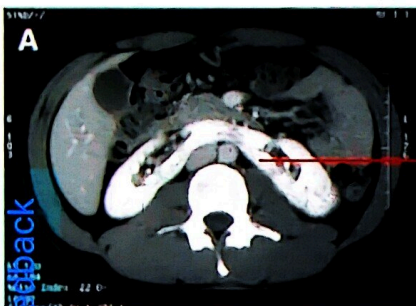
Dual transplantation : Renal + Hepatic.

Horseshoe Kidney

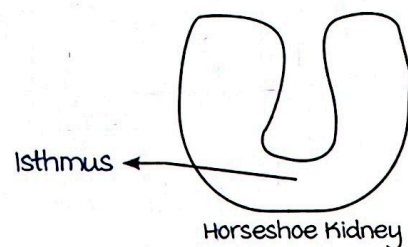
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- Defect : Fusion of lower poles of kidney.
- Genetic association : Turner's syndrome (30%).

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Horseshoe Kidney on CT scan



Horseshoe Kidney

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Complications: **Ureteral obstruction**

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- Hydronephrosis
- Urinary tract infection (UTI)
- Stone
- ↑ risk of Wilms' tumor : (4 fold)

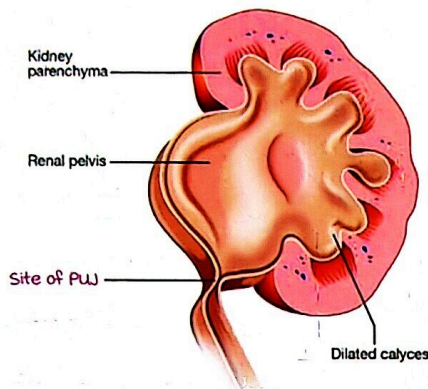
Obstructive Uropathy

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Anomalies causing obstructive uropathy.

Pelviureteric junction (PUJ) obstruction > Posterior urethral valve (PUV).

PUJ OBSTRUCTION :



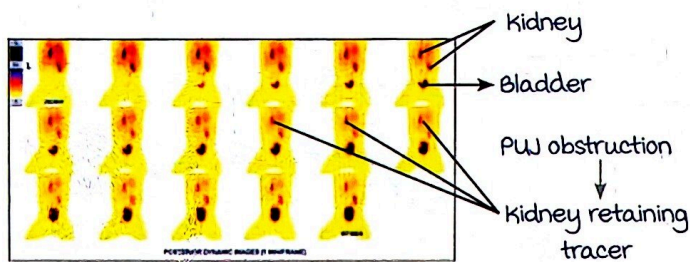
- Congenital.
- Usually unilateral : L > R
- Antenatal period : ↑ size of kidney (Hydronephrosis)
- Postnatal period : Abdominal mass, Stone, UTI.

Older child : **Dietl's crisis** : Intermittent episodes of severe abdominal pain

Nausea $\triangle$ Vomiting

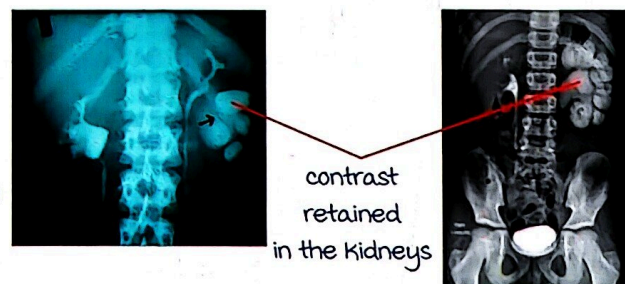
Investigations :

1. DTPA scan :

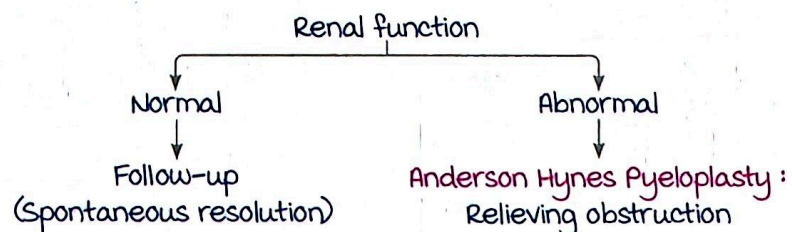


- Helps in assessment of renal function
→ management of PUJ obstruction.

2. Intravenous urography



management :



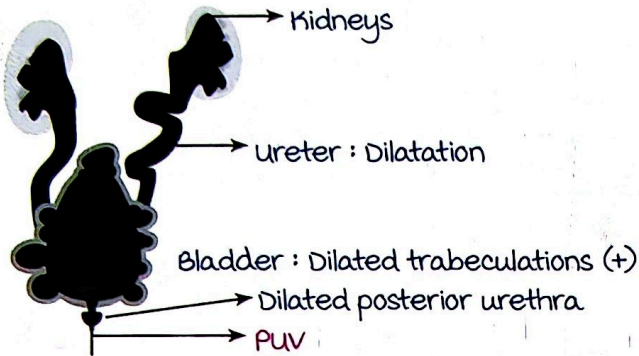
PUV :

- Only seen in male child.
- m/c severe cause of obstructive uropathy.

Pathophysiology :

- membrane valve causing obstruction.

Location : **Prostatic urethra** (Lower part) → Just distal to verumontanum.

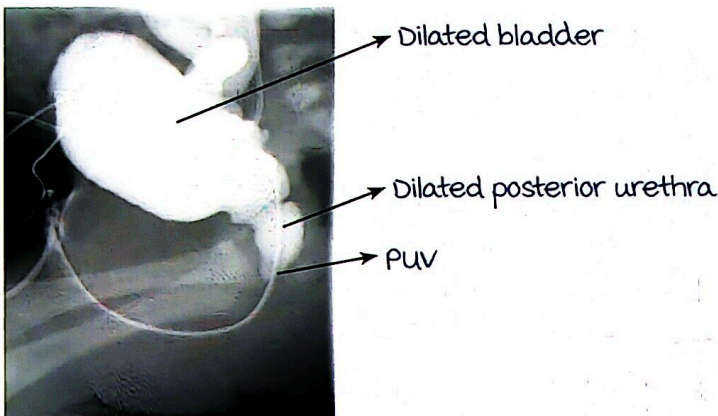


Clinical Features :

- Poor urine stream.
- Abdominal mass.
- UTI.
- Hydronephrosis.

Investigations :

mcu :



management :

Primary

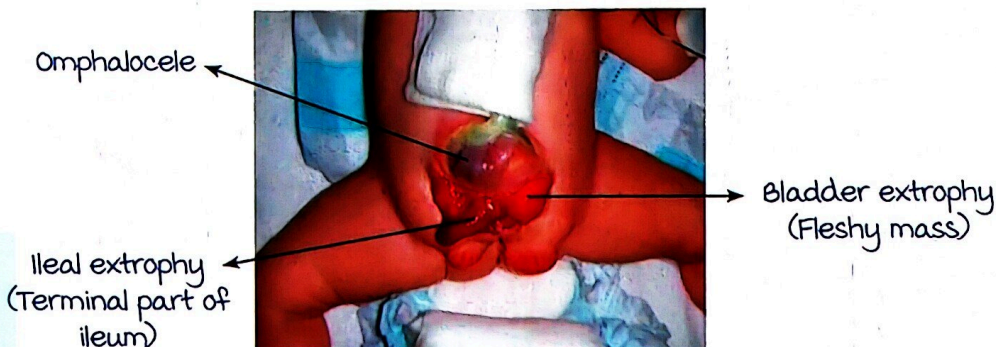
- Catheter → Relieve obstruction.

Definitive

- Endoscopic (Cystoscopic) fulguration of valve.

Cloacal Extrophy

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Feedback